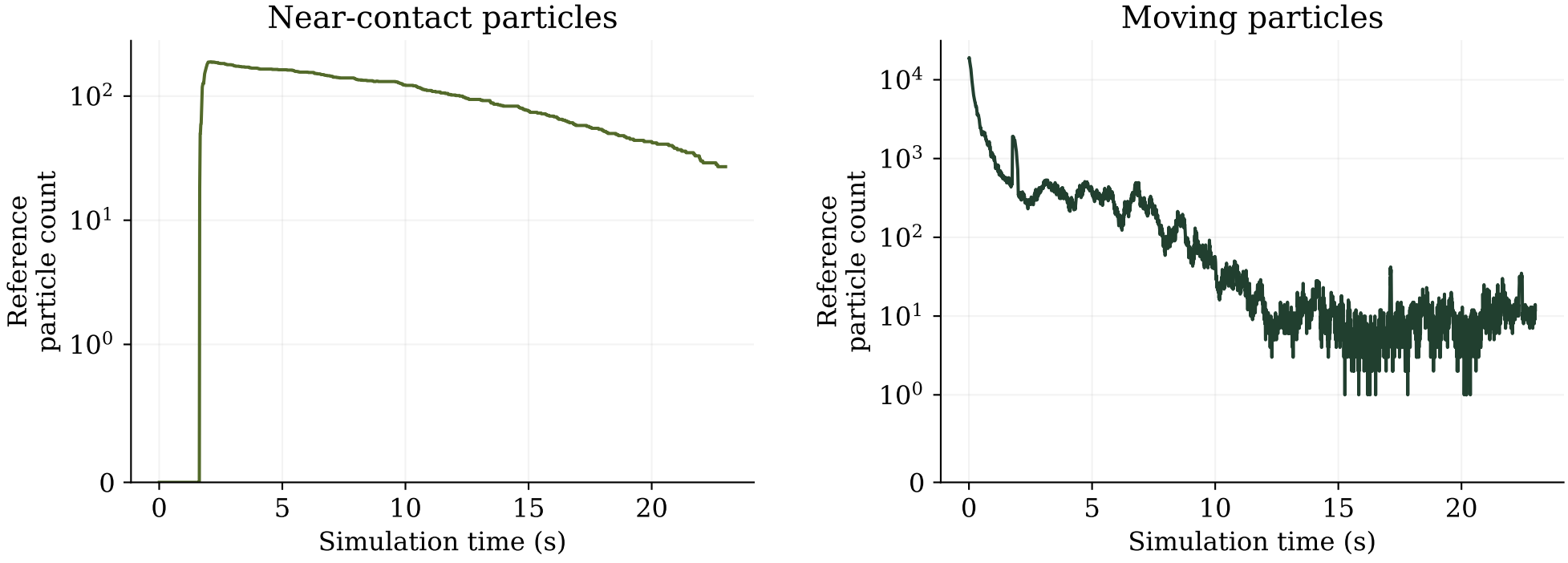

Figure 6: Reference populations in one paired rollout. Left: particles within 0.05 m of the body surface; right: particles moving faster than 0.01 m/s, using position-difference velocities at 200 Hz. Membership is recomputed at each step. Zero count makes the corresponding Figure 1 particle-position error undefined. The count axis is linear from 0 to 1 and logarithmic above 1; no smoothing or interpolation is applied.